

Gym Training Optimization Report

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1 Introduction

The project consists in a optimizing a Gym Training Plan solving an optimization problem which has been modeled and implemented in Pyomo with Gurobi as the solver. The output of the model is a structured 28-day training calendar that specifies, for each day, which muscle groups are trained, which exercises are selected, and how training volume is distributed. The document shows the meaning of the objective function, the structure of the constraints, and the modeling decisions that ensure interpretability and consistency. The objective of the project is showing how optimization techniques can be applied to a realistic decision-making problem in sports training.

2 Optimization Problem Description

2.1 Objective

The model aims to optimize the work of a personal trainer given certain objectives that must be completed by the client. The result of the model is a personalized 28-day training calendar.

It is crucial for an unexperienced client to be told the amount of volume necessary to avoid overtraining but maximizing muscle gain. To ensure this the model implements constraints that ensure biological recovery of the muscles worked, assigning certain rest days in the calendar. With this implementation we reduce highly the risk of injury which is also really important.

The model also considers the fact that certain muscles are fatigued when working other muscle groups, for example, when training chest, the triceps also works as a secondary muscle to move the weight. This is also considered as partial training to complete the objectives.

Knowing that most people have a limited schedule to work out, the model also limits the number of days available to work out to design a calendar that suits best subject to the availability of the client. There is also a maximum session duration which can be modified to adjust better to the needs of the client.

Scientific research supports the fact that including a compound movement enhances overall muscle growth and strength. The model considers this research and sets a daily basic exercise requirement.

3 Model Components and Logical Interpretation

3.1 Decision Variables

- **Training Day** (x_d): Binary variable indicating if training occurs on day d .
- **Sets Assigned** ($s_{d,e}$): Integer variable ≥ 0 representing the number of sets assigned on day d for exercise e .
- **Muscle Activation Day** ($w_{d,m}$): Binary helper variable indicating if muscle group m is worked on day d .
- **Positive/Negative Deviations** (δ_m^+, δ_m^-): Non-negative integers measuring the gap between actual and target monthly workload.

3.2 Logic and Big-M Formulation

Because the model needs to “switch off” certain behaviors (like recovery timers) when a muscle is not trained, we use **Big-M** notation. This allows the linear solver to handle conditional logic: if a binary variable is 0, the associated constraint is relaxed; if it is 1, the constraint is strictly enforced.

4 Mathematical Model Formulation

4.1 Sets

- E : Set of all exercises in the dataset.
- M : Set of muscle groups (Neck, Calves, Back, Chest, Shoulder, etc.).
- $D = \{1, \dots, 28\}$: Set of days in the training cycle.

4.2 Parameters

- t_e : Duration (minutes) per set of exercise e .
- $g_{e,m}$: Impact factor of exercise e on muscle m (2 for Main, 1 for Secondary, 0 otherwise).
- S_m : Target total workload (series) for muscle m over the month.
- r_m : Minimum rest days required after training muscle m .
- $T_{\max}$: Maximum duration of a single session (45 minutes).
- $N_{\max}$: Maximum allowed training days per month (12).
- Q_e : Quality score of exercise e .
- B_e : Binary indicator if exercise e is a “Basic” exercise ($Utility = 1$).
- $S_{\max}$: Maximum number of sets of the same exercise a day.

4.3 Constraints

- **Monthly Training Frequency Limit**

Since time is a limited resource, we must define the maximum number of days per month the user is willing or able to attend the gym. In the code, this is set to 12 days. This constraint prevents the solver from simply assigning training sessions every single day to meet the workload targets, forcing it to find the most efficient distribution of exercises within a limited calendar.

$$\sum_{d \in D} x_d \leq N_{\max}$$

- **Maximum Session Duration**

This constraint ensures that the total duration of all sets performed on a specific day does not exceed the limit of 60 minutes.

$$\sum_{e \in E} s_{d,e} \cdot t_e \leq x_d \cdot T_{\max}, \quad \forall d \in D$$

This formulation utilizes a **Big-M** logical structure. Crucially, because the right side of the inequality is multiplied by the binary decision variable x_d , the constraint serves two purposes:

- **Time Capping:** When $x_d = 1$ (the day is selected for training), the total time must be ≤ 60 minutes.
- **Logical Coupling:** When $x_d = 0$ (a rest day), the right side becomes zero. Since set assignments $s_{d,e}$ and durations t_e are non-negative, this mathematically forces all sets on that day to be zero.

Since our time is limited, this ensures the optimizer only allocates effort to the most valuable days while strictly respecting our 45-minute window.

- **Monthly Muscle Workload Balance**

This constraint calculates the total accumulated volume for each muscle group across the entire 28-day period and compares it to the target monthly series S_m .

$$\sum_{d \in D} \sum_{e \in E} (s_{d,e} \cdot g_{e,m}) + \delta_m^- - \delta_m^+ = S_m, \quad \forall m \in M$$

- **Muscle Activation Linking**

This is a logical bridge constraint that identifies which specific muscle groups are being targeted on any given day.

$$\sum_{e \in E} s_{d,e} \cdot g_{e,m} \leq M \cdot w_{d,m}, \quad \forall d \in D, \forall m \in M$$

- **Mandatory Muscle Recovery Period**

This constraint ensures that the user allows for sufficient physiological repair after targeting a specific muscle group.

$$\sum_{i=1}^{r_m} \sum_{e \in E} (s_{\text{idx},e} \cdot g_{e,m}) \leq (1 - w_{d,m}) \cdot M, \quad \forall d \in D, \forall m \in M$$

where $\text{idx} = (d + i - 1 \bmod 28) + 1$.

- **Exercise Repetition Limit**

To ensure variety and prevent repetitive strain, the number of sets for any specific exercise on a single day is capped at 3.

$$s_{d,e} \leq S_{\max}, \quad \forall d \in D, \forall e \in E$$

- **Training Density (Sliding Window)**

Ensures that within any 4-day period, the user does not train more than 3 times, guaranteeing at least one full rest day in every short block.

$$\sum_{k=d}^{d+3} x_k \leq 3, \quad \forall d \in \{1, \dots, 25\}$$

- **Daily Basic Exercise Requirement**

Mandates that every training session must include at least one “Basic” exercise, ensuring the core of the routine is high-utility.

$$\sum_{e \in E} B_e \cdot s_{d,e} \geq x_d, \quad \forall d \in D$$

5 Refined Objective Function

5.1 Mathematical Formulation

To enhance the model’s utility, we transition from a pure goal-programming approach to a multi-objective formulation that incorporates exercise quality (Q_e). This ensures the solver doesn’t just meet volume targets, but does so using the most effective movements available.

$$\min Z = \sum_{m \in M} (\delta_m^+ + \delta_m^-) - \sum_{d \in D} \sum_{e \in E} \left(\frac{s_{d,e} \cdot Q_e}{15 \times 12} \right)$$

5.2 Interpretation

The objective function balances two competing goals:

- **Target Accuracy:** The first term minimizes the absolute deviation from monthly muscle workload goals.
- **Quality Density:** The second term subtracts (effectively maximizes) the cumulative quality of the selected exercises.

5.3 Methodological Note

It is important to note that the detailed performance analysis and results discussed in the subsequent sections are based on the **previous model**. This decision was made because we could not identify a sufficiently objective or scientifically validated data source to assign quality scores (Q_e).

6 Conclusion

The session–duration plot (Figure 1) shows that nearly all training days lie at or extremely close to the imposed upper bound of 60 minutes, with only minor slack on Days 7 and 15. This indicates that the optimization is time-saturated. Which is a good sign as we want to have full training days to not waste time.

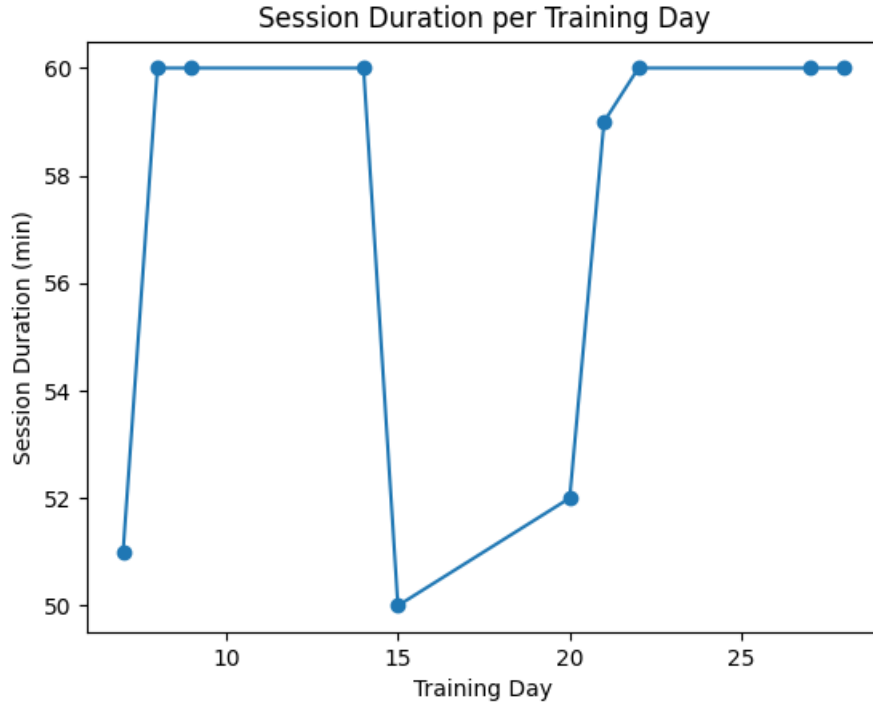


Figure 1: Training session duration per training day.

At the same time, the muscle–deviation plot (Figure 2) reveals that eight out of ten muscle groups reach their target volumes exactly. Only two deviations appear: a surplus of +8 sets in the shoulders and a deficit of −6 sets in the chest.

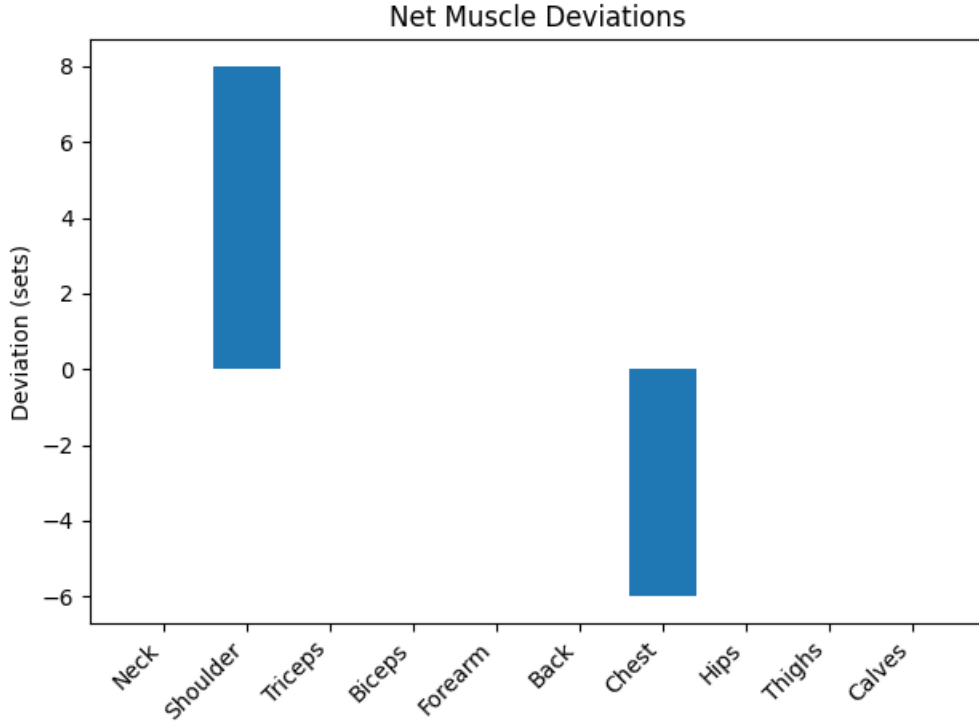


Figure 2: Net deviation from target volume by muscle group.

The internal logic of the solution becomes clearer when examining an individual training day. For example, Day 7 consists of a specialized arm and calf session comprising five biceps exercises (13 total sets) and five calf exercises (13 total sets).

Taken together, the two figures and the detailed day-level composition demonstrate that the optimizer has produced a near-Pareto-optimal training plan: time is almost fully utilized, muscle targets are met with high precision, and the only systematic deviation reflects an unavoidable biomechanical interaction rather than a modeling error.

Example: Detailed Structure of Day 7 To illustrate how the optimized solution translates into a real training session, Table 1 presents the full exercise composition of Day 7. This day corresponds to a specialized *biceps and calves* workout, designed to maximize hypertrophy stimulus while respecting the global time and recovery constraints.

Exercise	Sets
Incline Curl	3
Concentration Curl	3
Preacher Curl (Stirrups)	3
Arm Curl	3
One-Hand Curl	1
Standing Calf Raise	3
45° Calf Press	3
Calf Extension	1
Seated Calf Press	3
45° Calf Raise	3
Total Biceps Sets	13
Total Calf Sets	13

Table 1: Exercise and set composition for Day 7 of the optimized training plan.

This structure demonstrates how the optimization model achieves both biomechanical diversity and volume precision within a single session, using multiple exercise variants to fully recruit the targeted muscle groups while remaining within the session duration limit. Lastly, a real data base was used that we cleaned and transformed to fit our purpose.